

# Cosine Similarity of Two Vectors

- A **document** can be represented by a bag of terms or a long vector, with each attribute recording the *frequency* of a particular term (such as word, keyword, or phrase) in the document

| <i>Document</i> | <i>teamcoach</i> | <i>hockey</i> | <i>baseball</i> | <i>soccer</i> | <i>penalty</i> | <i>score</i> | <i>win</i> | <i>loss</i> | <i>season</i> |
|-----------------|------------------|---------------|-----------------|---------------|----------------|--------------|------------|-------------|---------------|
| Document1       | 5                | 0             | 3               | 0             | 2              | 0            | 2          | 0           | 0             |
| Document2       | 3                | 0             | 2               | 0             | 1              | 1            | 1          | 0           | 1             |
| Document3       | 0                | 7             | 0               | 2             | 1              | 0            | 3          | 0           | 0             |
| Document4       | 0                | 1             | 0               | 0             | 1              | 2            | 0          | 3           | 0             |

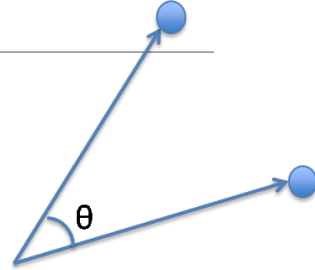
- Other vector objects: Gene features in micro-arrays
- Applications: Information retrieval, biologic taxonomy, gene feature mapping, etc.
- Cosine measure: If  $d_1$  and  $d_2$  are two vectors (e.g., term-frequency vectors), then

$$\cos(d_1, d_2) = \frac{d_1 \bullet d_2}{\|d_1\| \times \|d_2\|}$$

where  $\bullet$  indicates vector dot product,  $\|d\|$ : the length of vector  $d$

# Example: Calculating Cosine Similarity

□ Calculating Cosine Similarity:  $\cos(d_1, d_2) = \frac{d_1 \bullet d_2}{\|d_1\| \times \|d_2\|}$   $\text{sim}(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$



where  $\bullet$  indicates vector dot product,  $\|d\|$ : the length of vector  $d$

- Ex: Find the **similarity** between documents 1 and 2.

$$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0) \quad d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$$

- First, calculate vector dot product

$$d_1 \bullet d_2 = 5 \times 3 + 0 \times 0 + 3 \times 2 + 0 \times 0 + 2 \times 1 + 0 \times 1 + 0 \times 1 + 2 \times 1 + 0 \times 0 + 0 \times 1 = 25$$

- Then, calculate  $\|d_1\|$  and  $\|d_2\|$

$$\|d_1\| = \sqrt{5 \times 5 + 0 \times 0 + 3 \times 3 + 0 \times 0 + 2 \times 2 + 0 \times 0 + 0 \times 0 + 2 \times 2 + 0 \times 0 + 0 \times 0} = 6.481$$

$$\|d_2\| = \sqrt{3 \times 3 + 0 \times 0 + 2 \times 2 + 0 \times 0 + 1 \times 1 + 1 \times 1 + 0 \times 0 + 1 \times 1 + 0 \times 0 + 1 \times 1} = 4.12$$

- Calculate cosine similarity:  $\cos(d_1, d_2) = 25 / (6.481 \times 4.12) = 0.94$